



Modelling Financial Markets

Chapter 9: Realized Volatility



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Acronyms used in this material

- ▣ AR: AutoRegressive (model)
- ▣ ARFIMA: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ BV: Bipower Variation
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ DM: Diebold–Mariano (test)
- ▣ ETF: Exchange-Traded Fund
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-CJ: HAR model with Continuous and Jump components
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ MAE: Mean Absolute Error
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RK: Realised Kernel
- ▣ RQ: Realised Quarticity
- ▣ RV: Realised Variance
- ▣ SE: Standard Error
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

Outline

Motivation

Realised Variance

Microstructure Noise

Jumps

Stylised Facts of Realised Volatility

The HAR-RV Model

Forecasting: HAR vs GARCH

Rough Volatility

Summary

References

Volatility Is Latent: Can We Measure It?

The problem

- ▣ Volatility drives risk (Chapters 7–8), option prices (Chapter 12) and portfolio weights (Chapter 4)
- ▣ But it is never observed: one daily return is one draw from a distribution whose width we want
- ▣ GARCH (Chapter 5) **infers** today's variance from past daily returns through a model

The idea of this chapter

- ▣ Inside one day there are hundreds of price changes: add up their squares
- ▣ The sum **measures** the day's variance almost without a model: realised variance [Andersen & Bollerslev \(1998\)](#)
- ▣ Measured volatility can then be modelled and forecast like any time series
- ▣ Early idea: [Merton \(1980\)](#) showed that variance, unlike the mean, is estimated better with more frequent data

Realised Volatility: From Skeptics to Standard



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia
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- In the 1990s, GARCH forecasts seemed poor: they explained only a few % of squared daily returns
 - ▶ Andersen & Bollerslev (1998): the squared daily return is a very noisy proxy; against realised variance the same forecasts look good
- Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001): the distribution of realised volatility for currencies and stocks
- Andersen, Bollerslev, Diebold & Labys (2003): model realised volatility directly and forecast it
- Textbook treatment: Franke, Härdle & Hafner (2019)

Data Used in This Chapter

Series	Frequency	Period	Days
SPY (ETF on the S&P 500)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour UTC days	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- Up to 18 September 2026; returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %; a bar is labelled by its start time
- SPY: 78 returns per full day, 42 on half-days; bars off the 5-minute grid removed; missing bars inside a day carry the last price
- Bitcoin: 288 returns per day; days with fewer than 95% of the bars dropped (88 of 747 calendar days, mostly a data gap in October 2025)

Notation

- Day t is split into M intervals of length $\Delta = 1/M$; intraday returns $r_{t,1}, \dots, r_{t,M}$
 - ▶ SPY: $M = 78$ (5 minutes over 6.5 hours); Bitcoin: $M = 288$ (5 minutes over 24 hours)
- Daily return $r_t = \sum_{i=1}^M r_{t,i}$ (log returns add up)
- Daily variances in $\%^2$; annualised volatility $= \sqrt{252 \times \text{variance}}$ for SPY, $\sqrt{365 \times \text{variance}}$ for Bitcoin
- Overnight return (SPY): from the close to the next open; it is not inside the intraday bars

A Model for Intraday Prices

- Log price in continuous time: $dp(s) = \mu(s) ds + \sigma(s) dW(s)$, with W a Brownian motion (Chapter 11)
 - ▶ $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day
- **Integrated variance** of day t : the quantity we want to measure
 - ▶ $IV_t = \int_{t-1}^t \sigma^2(s) ds$
- Given the path of σ , the daily return is Normal with variance IV_t : $r_t | \sigma \sim N(\int \mu, IV_t)$
- IV_t is the natural daily variance; GARCH's σ_t^2 is its forecast from yesterday's information

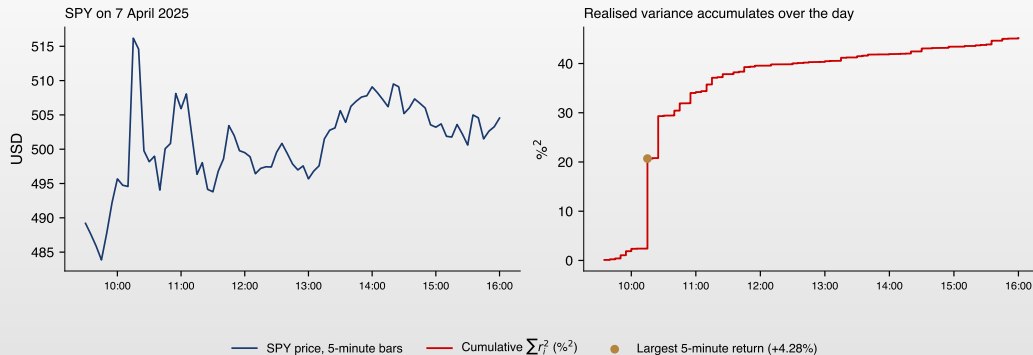
Definition: Realised Variance

- **Realised variance:** the sum of squared intraday returns

- ▶
$$RV_t = \sum_{i=1}^M r_{t,i}^2, \text{ realised volatility} = \sqrt{RV_t}$$

- As $M \rightarrow \infty$, $RV_t \rightarrow IV_t$: the drift vanishes and the sum of squares converges to the **quadratic variation** Barndorff-Nielsen & Shephard (2002)
 - ▶ The mean of SPY 5-minute returns is below 1% of their standard deviation: it can be ignored
- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$: unbiased but extremely noisy

One Day, 78 Returns: 7 April 2025

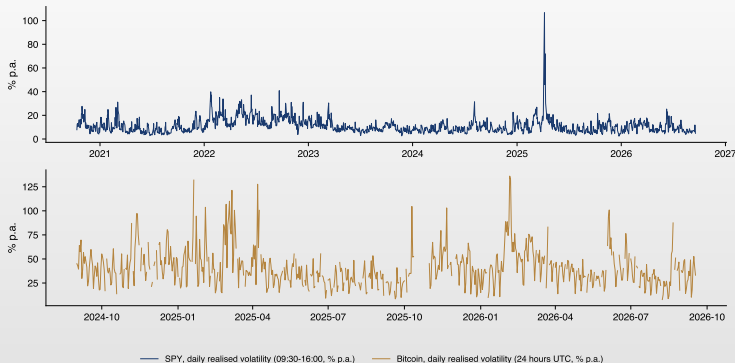


□ Left: SPY on 5-minute bars on the most volatile day of the sample; right: running sum of squared returns

Reading the Chart: 7 April 2025

- ▣ SPY opened -3.27% below the previous close, after tariff announcements; 78 five-minute returns followed
 - ▶ $RV = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- ▣ One 5-minute return of $+4.28\%$ (interval ending 10:15) contributed 41% of the day's realised variance
- ▣ Open-to-close the day ended $+3.09\%$: a small net move hides an extremely volatile path
- ▣ The daily return alone would say “a normal day”; realised variance says “one of the most turbulent days since 2020”

Daily Realised Volatility: SPY and Bitcoin

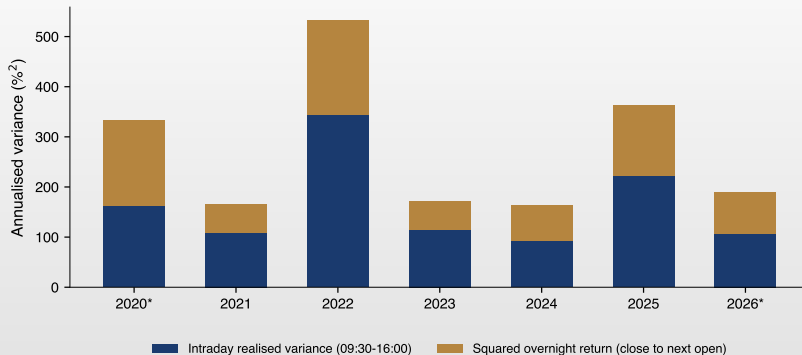


□ Annualised, from 5-minute returns; SPY: regular US session; Bitcoin: 24 hours; the gap in October 2025 is missing data

Reading the Realised Volatility Series

- SPY: average 12.9% p.a., median day 9.7%, range 2.4% to 106.7%
 - ▶ Mean above median: a few very volatile days pull the average up
- Bitcoin: average 45.1% p.a., median 37.8%, maximum 136.1% on 2026-02-05
- Both series show **clustering**: calm and turbulent weeks come in runs, as with GARCH (Chapter 5)
- Unlike GARCH's σ_t , every point here is a measurement, not a model output

Where Is the Risk: Day or Night?



- SPY, mean daily variance by year, annualised: intraday realised variance plus squared overnight return; 2020 and 2026 are partial years

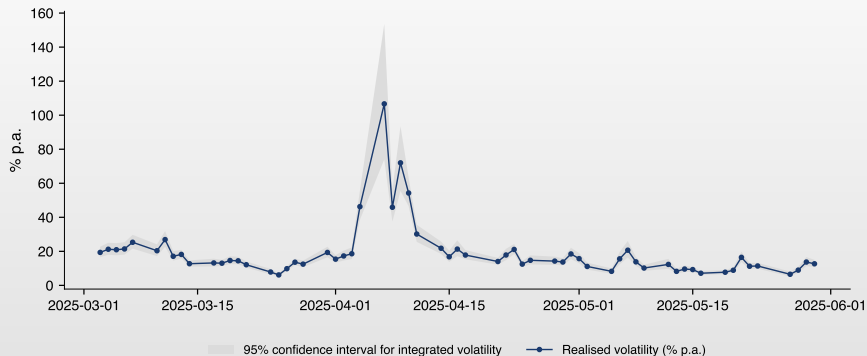
Reading the Overnight Decomposition

- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
 - ▶ News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised: close-to-close 16.6%; open-to-close 13.6%; overnight 10.1%
- Largest overnight move: -4.08% on 2024-08-05
- For daily risk we use **total** variance: $RV_t + r_{\text{overnight},t}^2$ (annualised mean 16.4%)
 - ▶ Alternatives: scale intraday RV up, or weight the two parts optimally [Hansen & Lunde \(2005\)](#)

How Precise Is Realised Variance?

- Central limit theorem for RV [Barndorff-Nielsen & Shephard \(2002\)](#): $\sqrt{M}(\text{RV}_t - \text{IV}_t) \rightarrow N(0, 2\text{IQ}_t)$
 - ▶ Integrated quarticity $\text{IQ}_t = \int \sigma^4(s) ds$, estimated by realised quarticity $\text{RQ}_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Standard error: $\text{se}(\text{RV}_t) = \sqrt{\frac{2}{3} \sum_i r_{t,i}^4}$
- 95% CI on the log scale (always positive): $\text{RV}_t \exp(\pm 1.96 \text{se}(\text{RV}_t)/\text{RV}_t)$
 - ▶ With constant volatility, $\text{se}/\text{RV} \approx \sqrt{2/M}$: 16% for $M = 78$
- RV is a measurement **with error**: it matters when RV is the target of a forecast (Section 6)

Confidence Intervals for Daily Volatility

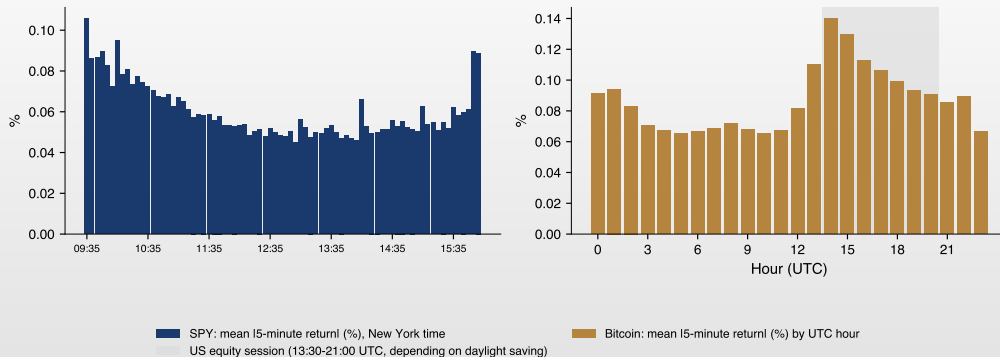


□ SPY, March–May 2025; band: 95% confidence interval for $\sqrt{252 IV_t}$ from realised quarticity

Reading the Confidence Intervals

- Median relative width of the interval (for volatility): 37% of its level
- 9 April 2025 (the day the tariff pause was announced): volatility 72%, CI [55%, 93%]
 - ▶ The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- Still far more precise than r_t^2 , whose relative error is of order $\sqrt{2} \approx 141\%$ under the Normal distribution
- Practical rule: trust the level of a single day to about $\pm 20\%$, and averages over weeks much more

Intraday Seasonality



Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day (full days only); right: Bitcoin by UTC hour

Reading the Seasonality Pattern

- SPY: a **U-shape**: 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes
Andersen & Bollerslev (1997)
 - ▶ The open prices overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC, when the US equity session opens; quietest at 10:00 UTC (ratio 2.2)
- RV sums over the whole day, so seasonality does not bias it; it matters for intraday risk and for sampling schemes

Recap: Realised Variance

- ▣ $RV_t = \sum_i r_{t,i}^2$ measures integrated variance without a model; precision improves with M
- ▣ Its standard error comes from realised quarticity: about $\pm 20\%$ for one day with 5-minute data
- ▣ For SPY, overnight returns add 38% of daily variance: use total variance for daily risk
- ▣ Volatility follows a daily U-shape (SPY) and a UTC-hour pattern (Bitcoin)

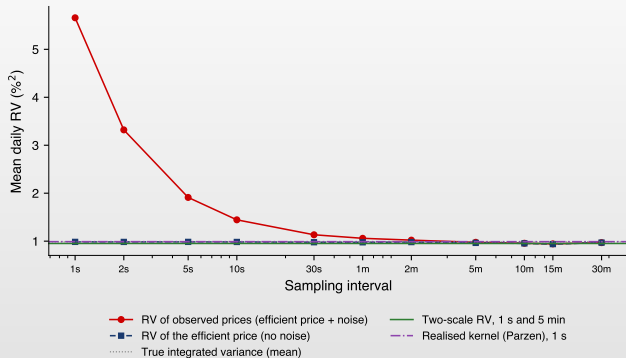
Why Not Use Every Tick?

- Observed prices are not the efficient price: $p_i^{\text{obs}} = p_i^* + \varepsilon_i$
 - ▶ Sources of ε : bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)
- With i.i.d. noise of variance ω^2 and n returns per day [Zhang, Mykland & Aït-Sahalia \(2005\)](#), [Hansen & Lunde \(2006\)](#)
 - ▶ $\mathbb{E}[\text{RV}_t^{(n)}] = \text{IV}_t + 2n\omega^2$: the bias **grows** with the sampling frequency
 - ▶ Observed returns become negatively autocorrelated: $\text{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: more returns reduce sampling error but add noise bias

Worked Example: the Size of the Noise Bias

- Daily IV = $1\%^2$ (16% p.a.), noise standard deviation $\omega = 0.01\%$ (one basis point), 6.5-hour session
- Bias $2n\omega^2$
 - ▶ 5 minutes ($n = 78$): $2 \times 78 \times 0.0001 = 0.016\%^2$, i.e. 1.6% of IV
 - ▶ 1 minute ($n = 390$): $0.078\%^2$ (7.8%)
 - ▶ 1 second ($n = 23\,400$): $4.680\%^2$: RV is almost six times IV
- MSE-optimal number of returns [Bandi & Russell \(2008\)](#): $n^* = (\text{IQ}/(4\omega^4))^{1/3}$
 - ▶ With $\text{IQ} = \text{IV}^2 = 1$: $n^* = 292$, one return every 80 seconds

The Volatility Signature Plot: Simulation



100 simulated days, one price per second; efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$

Reading the Simulated Signature Plot

- True mean IV = 0.98; RV of observed prices: 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
 - ▶ The bias matches $2n\omega^2$ from the previous slide
- Without noise, RV is flat in the sampling interval: the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
 - ▶ Error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045: using all data pays off only with a noise correction

Remedies: Sparse Sampling and Subsampling

- **Sparse sampling:** use one return every 5 minutes instead of every tick
 - ▶ Simple and robust; throws away most of the data [Liu, Patton & Sheppard \(2015\)](#)
- **Subsampling:** compute RV on K shifted grids (offsets $0, 1, \dots, K - 1$) and average
 - ▶ $RV_t^{\text{avg}} = \frac{1}{K} \sum_{k=0}^{K-1} RV_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV** [Zhang, Mykland & Aït-Sahalia \(2005\)](#): remove the remaining bias with the all-data RV
 - ▶ $TSRV_t = RV_t^{\text{avg}} - \frac{\bar{n}}{n} RV_t^{\text{all}}$, $\bar{n} = (n - K + 1)/K$; divide by $1 - \bar{n}/n$ in small samples

Realised Kernels

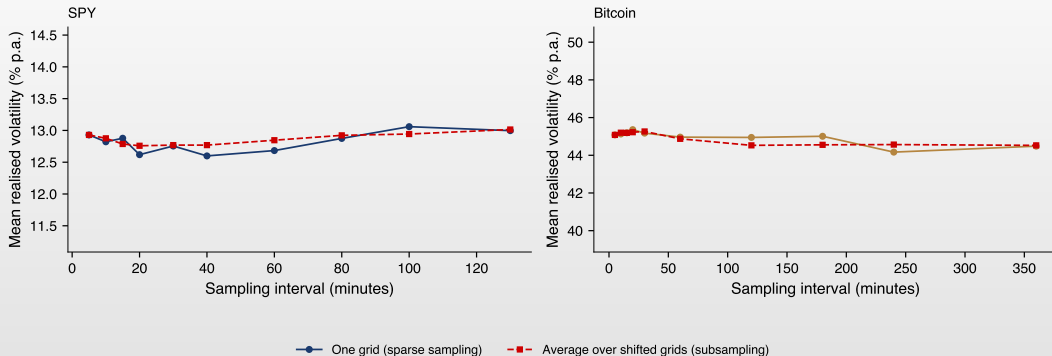


Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA
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- ▣ **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
 - ▶ $RK_t = \gamma_0 + 2 \sum_{h=1}^H k\left(\frac{h}{H+1}\right) \gamma_h$, $\gamma_h = \sum_i r_{t,i} r_{t,i-h}$
 - ▶ Parzen weight function k : 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- ▣ Noise makes γ_1 negative; the kernel adds it back and cancels the bias
- ▣ Bandwidth H grows with the noise-to-signal ratio: $H \approx 3.5 \xi^{4/5} n^{3/5}$, $\xi^2 = \omega^2 / \sqrt{IQ}$
- ▣ Shephard and co-authors developed kernels, bipower variation and the asymptotic theory of RV

The Signature Plot on Real Data



- Mean annualised RV by sampling interval: one grid vs the average over all shifted grids

Reading the Empirical Signature Plot

- SPY: 12.9% at 5 minutes, 12.8% at 30 minutes, 13.0% at 130 minutes: flat
 - ▶ First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Bitcoin: 45.1% at 5 minutes, 44.5% at 6 hours: flat as well
- On 5-minute SPY data: two-scale $RV / RV = 0.97$, realised kernel / $RV = 0.95$; correlations with RV 0.89 and 0.85
 - ▶ The corrections remove a bias that is not there and add estimation noise
- Consistent with [Liu, Patton & Sheppard \(2015\)](#): almost 400 estimators on 31 assets rarely beat 5-minute RV; noise-robust estimators pay off with tick data

Recap: Microstructure Noise

- Noise adds $2n\omega^2$ to RV: sampling too often overstates volatility
- The signature plot is the diagnostic: RV should not depend on the sampling interval
- Remedies: 5-minute sampling, subsampling, two-scale RV, realised kernels
- For liquid assets at 5 minutes (SPY, Bitcoin) plain RV is the benchmark

Prices Jump

- Add jumps to the model: $dp = \mu ds + \sigma dW + dJ$, with J a jump process (news, Chapter 11)
 - ▶ Quadratic variation now has two parts: $RV_t \rightarrow IV_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
- Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)
- We need an estimator of IV_t that ignores jumps

Bipower Variation



Ole E. Barndorff-Nielsen, 2007

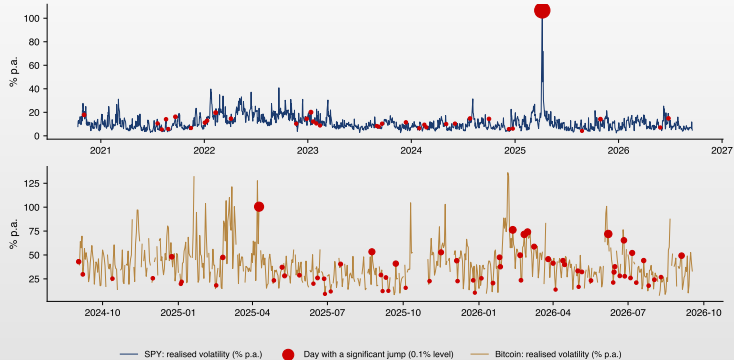
Photo: Thomas Steiner (2007); CC BY-SA 2.5;
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- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
 - ▶ $BV_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^M |r_{t,i}| |r_{t,i-1}| \rightarrow IV_t$
 - ▶ $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0, 1)$, so $\mu_1^{-2} = \pi/2 = 1.571$
- A jump enters one return: it is multiplied by a small neighbour, so its effect vanishes as M grows
- Jump part of the day: $\max(RV_t - BV_t, 0)$
- Barndorff-Nielsen (1935–2022) also shaped the Lévy-process models used for jumps

A Test for Jumps

- ▣ Ratio statistic [Huang & Tauchen \(2005\)](#), [Barndorff-Nielsen & Shephard \(2006\)](#):
$$z_t = \frac{(RV_t - BV_t)/RV_t}{\sqrt{\theta M^{-1} \max(1, TQ_t/BV_t^2)}}$$
 - ▶ $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $TQ_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
- ▣ Under no jumps $z_t \rightarrow N(0, 1)$; a jump makes $RV > BV$ and z_t large and positive
- ▣ One test per day: at level α we expect $\alpha \times$ days false alarms
 - ▶ We use $\alpha = 0.1\%$ (critical value 3.09): about 1.5 false jump days for SPY
- ▣ Significant day: jump part $J_t = RV_t - BV_t$, continuous part $C_t = RV_t - J_t$; otherwise $J_t = 0$

Jump Days: SPY and Bitcoin



□ Red points: days with a significant jump at the 0.1% level; point size proportional to the jump part J_t

Reading the Jump Test

- SPY: 35 jump days of 1 489 (2.4%); jumps are 2.8% of total realised variance
 - ▶ On a jump day the jump is on average 35% of that day's RV; BV/RV averages 0.95 over all days
 - ▶ Largest statistic: 2021-08-27 ($z = 5.2$), jump in the 10:00–10:05 interval, when the Federal Reserve chair's Jackson Hole speech was released
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
 - ▶ Many small jumps on quiet days (weekends): one isolated move dominates a calm day
- The turbulent days of April 2025 are mostly **not** jump days: many large returns in a row raise BV as well

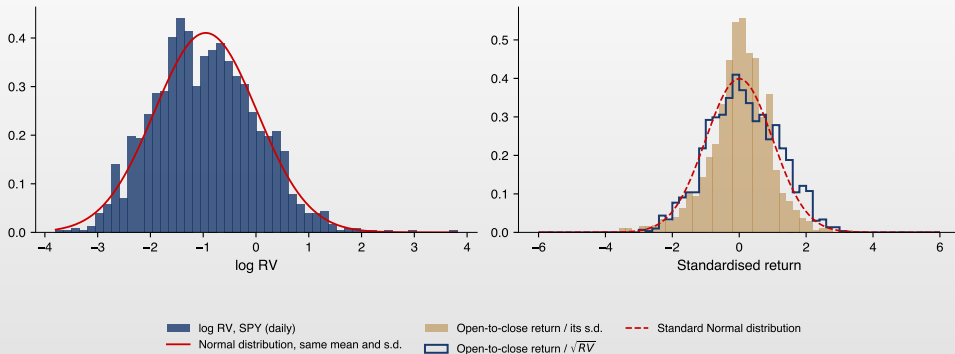
Recap: Jumps

- RV measures total quadratic variation: diffusion plus squared jumps
- Bipower variation is robust to jumps; $RV - BV$ estimates the jump part
- The ratio test flags about 2.4% of SPY days at the 0.1% level; jumps are a small share of total variance
- Test many days, expect false alarms: at 5% the test flags 235 SPY days versus 74 expected under no jumps
 - ▶ Jumps are separated from continuous variation in the HAR-CJ model (Section 5, Seminar 9)

What Realised Volatility Revealed

- Once volatility is measured, its properties can be studied directly Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001)
 - ▶ (1) Realised variance is very skewed, but its logarithm is close to Normal
 - ▶ (2) Returns divided by realised volatility are close to standard Normal
 - ▶ (3) Log realised volatility has **long memory**: autocorrelations decay slowly
 - ▶ (4) Implied volatility exceeds subsequent realised volatility: the variance risk premium
- Each fact shapes the model of Section 6: HAR on log RV

Distributions: log RV and Standardised Returns

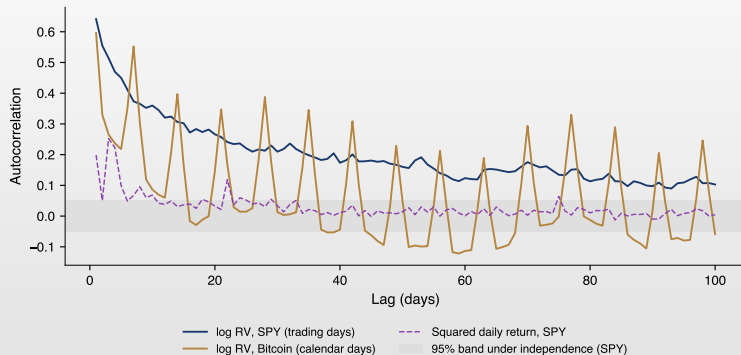


- Left: daily log RV of SPY and the Normal density; right: open-to-close returns scaled by their standard deviation (filled) and by $\sqrt{RV_t}$ (outline)

Reading the Distributions

- SPY RV: skewness 20.0, kurtosis 550; log RV: skewness 0.29, kurtosis 3.20 (Normal: 0 and 3)
 - ▶ Bitcoin log RV: skewness -0.27 , kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73; divided by $\sqrt{RV_t}$: 2.65, standard deviation 1.05
 - ▶ Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns (Chapter 1) come mostly from changing volatility: returns are a **mixture** of Normal distributions with variances IV_t

Long Memory in Realised Volatility

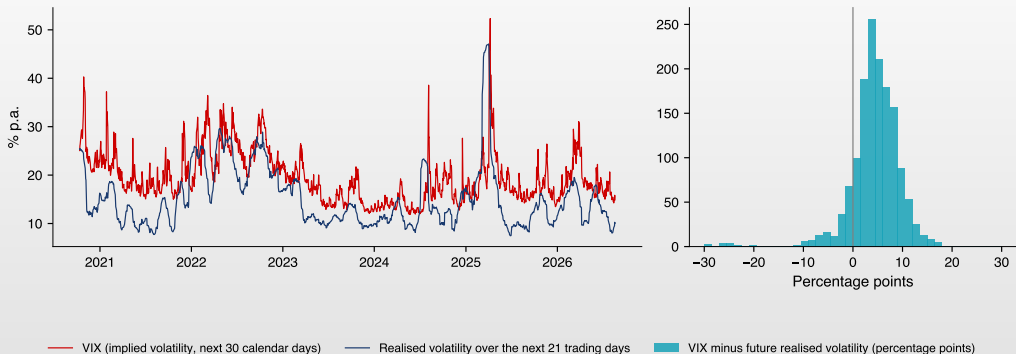


- Autocorrelations up to 100 days: log RV of SPY (total variance) and Bitcoin, squared daily SPY returns

Reading the Autocorrelations

- SPY log RV: 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days: **long memory**
 - ▶ An AR(1) would decay geometrically: $0.64^{22} \approx 0.00005$
- Squared daily returns show the same persistence, hidden by noise: 0.20 at lag 1
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
 - ▶ Weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.): no equity or macro news, fewer institutional traders

Implied vs Realised Volatility



- Left: VIX and SPY realised volatility over the next 21 trading days (including overnight returns); right: the gap between them

Reading the Variance Risk Premium

- VIX: the market's 30-day volatility, implied by S&P 500 options (Chapter 12); mean 19.5% vs 15.2% realised afterwards
 - ▶ VIX exceeded the subsequent realised volatility on 88% of 1 468 days; mean gap 4.4 percentage points
- **Variance risk premium:** investors pay to insure against volatility spikes Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
- As a forecast: regressing future variance on VIX^2 gives slope 0.56 (SE 0.05), $R^2 = 0.23$: VIX is informative but biased upwards
- The premium turns negative when a shock arrives after a calm period (March–April 2025)

Recap: Stylised Facts

- $\log RV$ is close to Normal: model and forecast volatility in logs
- Returns scaled by $\sqrt{RV}$ are close to $N(0, 1)$: fat tails come from changing volatility
- Long memory: the model must weigh the last month, not only yesterday
- Implied volatility carries a risk premium: it is an upward-biased forecast

Traders with Different Horizons

- Heterogeneous market hypothesis [Müller et al. \(1997\)](#): intraday traders, weekly hedgers and monthly asset managers all react to volatility
 - ▶ Each group looks at volatility over its own horizon: day, week, month
- Long-horizon volatility drives short-horizon volatility more than the reverse: a volatility cascade
- A sum of a few components with different persistence mimics long memory [Corsi \(2009\)](#)
- Alternative: fractionally integrated models (ARFIMA): harder to estimate, similar forecasts

The HAR-RV Model

- Corsi (2009): $RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
 - ▶ $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$ (week), $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$ (month)
 - ▶ Bitcoin trades every day: week = 7 days, month = 30 days
- An AR(22) with only 3 free slopes: weight $\beta_d + \beta_w/5 + \beta_m/22$ on lag 1, $\beta_w/5 + \beta_m/22$ on lags 2–5, $\beta_m/22$ on lags 6–22
- Estimation by OLS; errors are heteroskedastic and autocorrelated: use Newey–West (HAC) standard errors
 - ▶ log-HAR: the same regression for $\ln RV$; forecast $\exp(\hat{y} + \hat{s}^2/2)$ corrects the log bias

Extensions of HAR

- **HAR-CJ** Andersen, Bollerslev & Diebold (2007): separate continuous and jump parts as regressors
 - ▶ $\ln RV_{t+1} = \beta_0 + \beta_d \ln C_t + \beta_w \ln C_t^{(w)} + \beta_m \ln C_t^{(m)} + \beta_J \ln(1 + J_t) + \varepsilon_{t+1}$
- **HARQ** Bollerslev, Patton & Quaedvlieg (2016): less weight on yesterday's RV when it is imprecise (high realised quarticity)
 - ▶ β_d becomes $\beta_d + \beta_Q \sqrt{RQ_t}$ with $\beta_Q < 0$
- Weekend or holiday dummies, leverage (negative returns raise future RV), implied volatility as an extra regressor
- Multi-step: regress $\sum_{j=1}^h RV_{t+j}$ directly on the same regressors

HAR Estimates: SPY and Bitcoin

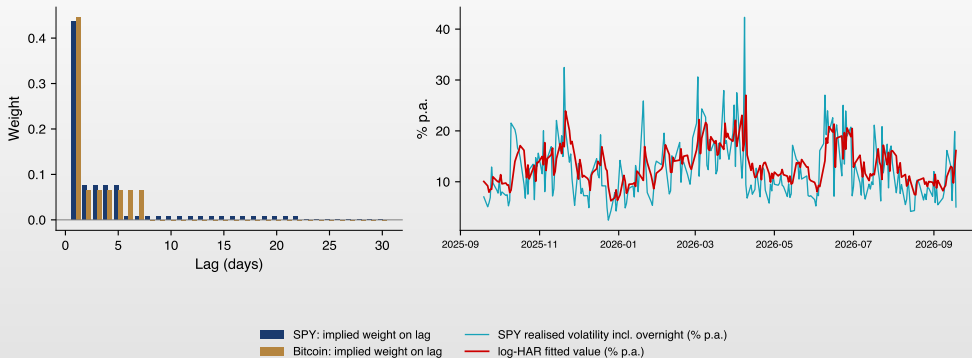
	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273 (0.064)	0.399 (0.089)	0.222 (0.096)	0.114 (0.053)	0.31
SPY, log-HAR	-0.079 (0.027)	0.362 (0.037)	0.337 (0.048)	0.167 (0.044)	0.47
Bitcoin, HAR (levels)	1.496 (0.412)	0.412 (0.089)	0.409 (0.122)	-0.065 (0.097)	0.34
Bitcoin, log-HAR	0.223 (0.094)	0.380 (0.041)	0.465 (0.080)	-0.022 (0.098)	0.36

- Full samples; SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days; Bitcoin: 24-hour RV, 547 days
- Newey–West standard errors in brackets

Reading the HAR Estimates

- SPY: all three components are significant; in logs $\hat{\beta}_d = 0.362$, $\hat{\beta}_w = 0.337$, $\hat{\beta}_m = 0.167$
 - ▶ Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH (Chapter 5)
- log-HAR fits better: $R^2 = 0.47$ vs 0.31 in levels, where a few extreme days dominate the squared errors
- Bitcoin: daily and weekly components matter; the monthly coefficient is close to 0 (-0.022 , SE 0.098)
 - ▶ Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

Implied Lag Weights and In-Sample Fit



Left: weights of log-HAR on each past day; right: SPY, last 250 days, realised volatility and log-HAR fitted values

Reading the Lag Weights

- SPY: yesterday gets 0.437, each of days 2–5 gets 0.075, each of days 6–22 gets 0.008
- The staircase decays slowly: the month still holds weight $22 \times 0.008 \approx 0.167$
 - ▶ Three coefficients reproduce the long memory of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise

Worked Example: A log-HAR Forecast

- Rounded SPY log-HAR: $\ln RV_{t+1} = -0.08 + 0.36 \ln RV_t + 0.34 \ln RV_t^{(w)} + 0.17 \ln RV_t^{(m)}$
- Today $RV_t = 2.0$, weekly average of logs $\ln 1.2$, monthly $\ln 0.9$ (all in $\%^2$)
 - ▶ $-0.08 + 0.36 \times 0.693 + 0.34 \times 0.182 + 0.17 \times (-0.105) = 0.214$
 - ▶ Forecast (ignoring the small variance correction) $e^{0.214} = 1.24\%^2$, i.e. $\sqrt{252 \times 1.24} = 17.7\%$ p.a.
- Mean reversion: tomorrow's forecast lies between today's 2.0 and the monthly level 0.9

Recap: HAR-RV

- HAR regresses tomorrow's RV on daily, weekly and monthly averages: OLS, three slopes
- It approximates long memory with a staircase of lag weights
- Estimate in logs; report Newey–West standard errors
- Extensions: jumps (HAR-CJ), measurement error (HARQ), calendar effects, implied volatility

Forecasting Design

- Target: tomorrow's daily variance; SPY: total variance, 2022-01-03 to 18 September 2026 (1 180 days); Bitcoin: 2025-03-01 onwards (422 days)
 - ▶ Every forecast uses only data available the evening before
- Five models
 - ▶ log-HAR and HAR on realised variance, re-estimated daily on an expanding window
 - ▶ GARCH(1,1) with Student- t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days [Bollerslev \(1986\)](#)
 - ▶ EWMA with $\lambda = 0.94$ (Chapter 5) and "yesterday's RV" (random walk)
- HAR sees intraday data; GARCH and EWMA see only daily returns

Evaluating Against a Noisy Target

- True variance is unobserved; we compare forecasts F_t with the proxy RV_t
 - ▶ A loss is **robust** if the ranking with a noisy unbiased proxy equals the ranking with true variance
Patton (2011)
- Two robust losses
 - ▶ $MSE = (RV_t - F_t)^2$: dominated by the most volatile days
 - ▶ $QLIKE = \frac{RV_t}{F_t} - \ln \frac{RV_t}{F_t} - 1 \geq 0$: depends only on the ratio; penalises under-prediction more
- Common mistake: MAE, or MSE on volatility instead of variance, is **not** robust

Tests: Diebold–Mariano and Mincer–Zarnowitz

- Diebold & Mariano (1995): loss difference $d_t = L(RV_t, F_t^A) - L(RV_t, F_t^B)$; $t = \bar{d}/\text{se}_{\text{NW}}(\bar{d}) \rightarrow N(0, 1)$
 - ▶ $t < 0$: model A has the lower loss; HAC standard error because d_t is autocorrelated (Chapter 8)
- Mincer & Zarnowitz (1969): $RV_t = a + b F_t + u_t$; an unbiased forecast has $a = 0$, $b = 1$
 - ▶ Joint Wald test with Newey–West covariance; R^2 measures how much of the variation the forecast explains
 - ▶ R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026

Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.310	–	4.15	1.16	0.30
HAR	0.326	–1.95 (0.052)	4.48	0.75	0.26
GARCH(1,1)- t	0.343	–2.30 (0.021)	4.87	0.70	0.20
EWMA	0.393	–4.02 (< 0.001)	5.14	0.74	0.14
Yesterday's RV	0.502	–5.95 (< 0.001)	5.41	0.54	0.29

□ 1180 one-day-ahead forecasts of total daily variance; DM: QLIKE differences, Newey–West; $t < 0$ favours log-HAR

Reading the SPY Results

- log-HAR has the lowest QLIKE and MSE: QLIKE 10% below GARCH(1,1)- t (DM $t = -2.30$, $p = 0.021$)
 - ▶ Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally ($p = 0.052$)
- MSE differences are not significant: a few crisis days (April 2025) dominate the squared errors
- Mincer–Zarnowitz: log-HAR $b = 1.16$ ($p = 0.554$ for $a = 0$, $b = 1$), GARCH $b = 0.70$: no significant bias, but log-HAR explains more (R^2 0.30 vs 0.20)
- Yesterday's RV has a high R^2 but $b = 0.54$: right direction, far too extreme

Results: Bitcoin, 2025–2026

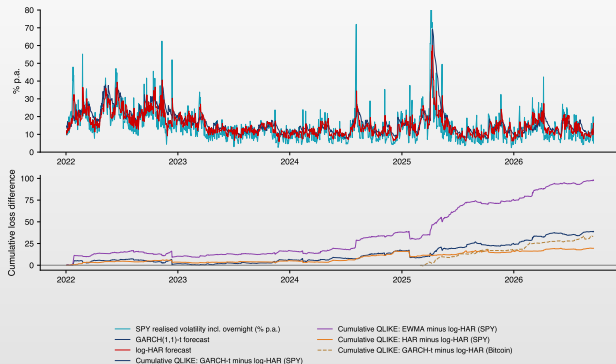
Model	QLIKE	DM vs log-HAR (p)	MSE	MZ b	MZ R^2
log-HAR	0.282	–	26.6	0.82	0.38
HAR	0.409	–1.22 (0.223)	27.6	0.89	0.34
GARCH(1,1)- t	0.360	–4.32 (< 0.001)	42.2	0.55	0.23
EWMA	0.346	–2.96 (0.003)	36.3	0.64	0.18
Yesterday's RV	0.466	–5.38 (< 0.001)	35.0	0.58	0.33

□ 422 one-day-ahead forecasts of 24-hour realised variance; same models and tests as for SPY

Reading the Bitcoin Results

- log-HAR wins again: QLIKE 22% below GARCH(1,1)- t (DM $t = -4.32$, $p < 0.001$)
 - ▶ GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves
- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360): after calm weekends it predicts too low a variance, and QLIKE punishes under-prediction
- Intraday information improves Bitcoin risk forecasts, as found for high-frequency entropy measures by [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Crypto-assets behave like a distinct asset class with their own volatility dynamics [Pele et al. \(2023\)](#)

Forecasts over Time

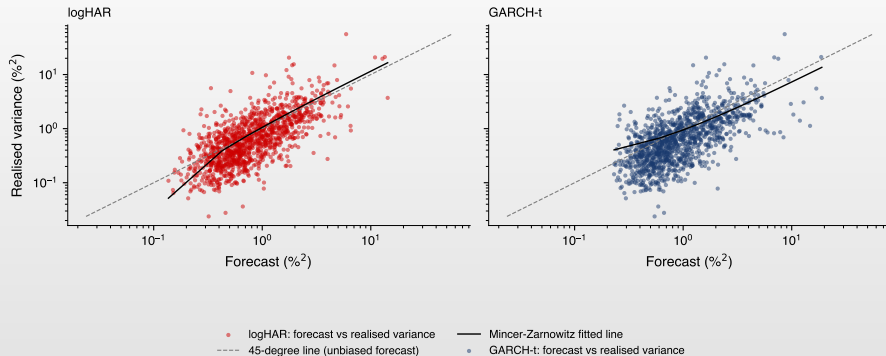


- Top: SPY realised volatility (with overnight) and the two forecasts, annualised; bottom: cumulative QLIKE loss of each model minus log-HAR (rising line = log-HAR better)

Reading the Forecast Paths

- After a shock both forecasts jump; GARCH stays high longer (its persistence is close to 1), log-HAR returns faster
 - ▶ April 2025: the GARCH forecast stayed above 40% p.a. until late April, while realised volatility was below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes: the gain is not driven by one day
- EWMA loses most: a fixed $\lambda = 0.94$ reacts too much to single days

Mincer–Zarnowitz Regressions



□ SPY, 2022–2026: realised variance against the forecast (log scales); dashed: 45-degree line; black: fitted regression in levels

Reading the Mincer–Zarnowitz Plots

- log-HAR forecasts span a wider range: they separate calm from turbulent days better
- Only 1% of GARCH forecasts are below $0.26\%^2$ per day (log-HAR: minimum $0.14\%^2$): on calm days realised variance falls far below GARCH
- The cloud is wide for both: one day's variance is hard to predict, and RV itself is measured with error
- Weekly horizon and model extensions: Seminar 9 (B6, B7)

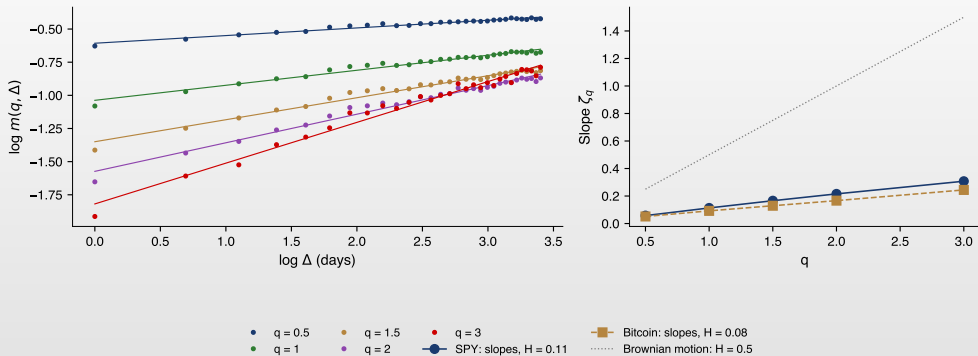
Recap: Forecasting

- Evaluate variance forecasts with robust losses (QLIKE, MSE on variance) and test differences with Diebold–Mariano
- Intraday data help: log-HAR beats GARCH(1,1)- t by 10% (SPY) and 22% (Bitcoin) in QLIKE
- Work in logs: HAR in levels loses under QLIKE when it under-predicts
- MSE rankings rest on a few crisis days and are rarely significant

Is Volatility Rough?

- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion** with Hurst exponent $H \approx 0.1$
 - ▶ $H = 0.5$: ordinary Brownian motion; $H < 0.5$: increments are negatively correlated, paths are rougher; $H > 0.5$: smoother, trending
- Scaling law: $m(q, \Delta) = \frac{1}{T} \sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q \propto \Delta^{qH}$
 - ▶ Regress $\ln m(q, \Delta)$ on $\ln \Delta$: slope $\zeta_q = qH$; then H from the slopes across q
- Rough models reproduce the shape of the implied volatility smile for short maturities (Chapter 12)

Scaling of Log Volatility Increments



□ Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \dots, 30$ days; right: slopes ζ_q against q

Reading the Roughness Estimates

- Straight lines on the log-log scale: the scaling law holds; slopes are linear in q
 - ▶ $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI $[0.09, 0.15]$), 0.08 for Bitcoin: far below 0.5
- Caution: RV measures volatility with error, and measurement error lowers $\hat{H}$
 - ▶ A simulated Brownian motion gives $\hat{H} = 0.46$; plus noise of a similar relative size, $\hat{H} = 0.18$ (Seminar 9, B8)
- Roughness and HAR agree in practice: both say that recent days matter much more than distant ones

Recap: Rough Volatility

- The moments of log-volatility increments scale like Δ^{qH} with $H \approx 0.1$
- Our data: $\hat{H} = 0.11$ (SPY) and 0.08 (Bitcoin)
- Part of the roughness can be measurement error in RV; read $\hat{H}$ as an upper bound on roughness
- Rough volatility models are an active research area in option pricing

Summary

- ▣ Realised variance turns intraday prices into a measurement of daily variance, with a known standard error
- ▣ Sample at 5 minutes for liquid assets; use noise-robust estimators only with tick data; add the overnight return for daily risk
- ▣ Bipower variation separates jumps from continuous variation; jumps are rare and a small share of variance
- ▣ log RV is nearly Normal and has long memory: the log-HAR model captures it with three coefficients
- ▣ Out of sample, log-HAR beats GARCH for SPY and Bitcoin under QLIKE; MSE comparisons are inconclusive

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars Tick data, or an illiquid stock (BVB)	RV at 5 minutes, subsampled Realised kernel or two-scale RV; check the signature plot
Daily risk of an exchange-traded asset Separate news jumps from diffusion Daily or weekly variance forecast	Total variance: RV plus squared overnight return Bipower variation, ratio test at 0.1% log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz

Coming Next: Market Microstructure (Chapter 10)

- Where does the noise in high-frequency prices come from? The limit order book, bid–ask spreads, tick sizes
- Liquidity and market impact: how prices react to trading
- High-frequency trading and crypto exchange order books
- Further reading: Andersen, Bollerslev, Diebold & Labys (2003), Corsi (2009), Franke, Härdle & Hafner (2019)

Further Study on Quantinar

Advanced courses on quantinar.com

-  Nonparametric and Semiparametric Models
-  Kalman Filter
-  Applied Time Series Analysis with Python

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

References (1/3)

- ▣ Andersen, T.G., Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3), 115–158.
- ▣ Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X. (2007). Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *Review of Economics and Statistics*, 89(4), 701–720.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2001). The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96(453), 42–55.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.
- ▣ Bandi, F.M., Russell, J.R. (2008). Microstructure noise, realized variance, and optimal sampling. *Review of Economic Studies*, 75(2), 339–369.
- ▣ Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society, Series B*, 64(2), 253–280.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.
- ▣ Barndorff-Nielsen, O.E., Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.

References (2/3)

- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.
- Bollerslev, T., Tauchen, G., Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.
- Carr, P., Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3), 1311–1341.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- Hansen, P.R., Lunde, A. (2006). Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2), 127–161.
- Huang, X., Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499.

References (3/3)

- ▣ Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- ▣ Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- ▣ Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic Forecasts and Expectations*, NBER, 3–46.
- ▣ Müller, U.A., Dacorogna, M.M., Davé, R.D., Olsen, R.B., Pictet, O.V., von Weizsäcker, J.E. (1997). Volatilities of different time resolutions: analyzing the dynamics of market components. *Journal of Empirical Finance*, 4(2–3), 213–239.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- ▣ Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411.